



AegisAlpha: Autonomous Multi-Agent Options Trading Desk

LabLab.ai × Alpaca AI Trading Agents Hackathon Submission | Team Returnee

Live URL: <https://aegis-alpha-desk.vercel.app> | GitHub: [DeathKnell837/band-of-agents-hackathon](https://github.com/DeathKnell837/band-of-agents-hackathon)

1. The Core Problem in AI Trading

Why 99% of LLM trading algorithms fail in production

Conventional Retail Bots	The AegisAlpha Paradigm
• Unhedged Risk: Place raw directional stock or naked options with unlimited loss potential. • Hallucinated Sizing: LLMs allocate arbitrary position sizes without mathematical risk bounding. • No Greeks Awareness: Blind to Delta convexity, Theta decay, and Implied Volatility crush.	• Defined-Risk Spreads Only: Executes Bull Call Spreads, Bear Put Spreads, and Iron Condors with fixed maximum loss. • Zero-LLM Risk Authority: 7 hardcoded Python gates enforce strict 2% equity risk caps (\$2,000 max). • Quantitative Greeks Optimizer: Real-time Alpaca chain filtering targeting optimal 0.40/0.20 Delta ratios.

2. Autonomous 4-Agent Dialectic Architecture

Separating hypothesis generation from mathematical execution

Agent	Core Engine & Role	Output / Mechanism
1. Market Scanner	Alpaca IEX Data Client	Ingests real-time 1m/5m/1d bars, calculates 30-day realized volatility & 5/20 MA momentum.
2. Alpha Strategist	Qwen-2.5-72B (Featherless AI)	Evaluates regime, scores thesis confidence, and formulates multi-leg spread structures.
3. Greeks Optimizer	Alpaca Options Historical Feed	Filters option contracts for strike expiration, liquidity, and 0.40/0.20 Delta ratios.

4. Risk Gatekeeper	Deterministic Risk Engine	7 hardcoded mathematical safety gates. Pass/Veto verdict with zero LLM override.
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3. The 7 Deterministic Risk Guardrails

Mathematical boundaries protecting portfolio equity (\$100k Account PA3PL5AZ85K6)

Gate	Name	Constraint Rule	Enforcement
Gate 1	Defined-Risk Only	No naked calls/puts. Spreads only.	HARD BLOCK
Gate 2	Max Risk per Trade	Max \$2,000 loss (2.0% equity cap)	VETO / RESIZE
Gate 3	Portfolio Options Cap	Max \$20,000 total active in options (20%)	HARD BLOCK
Gate 4	Bid-Ask Liquidity	Spread must be <15% of contract mid-price	SLIPPAGE FILTER
Gate 5	DTE Horizon	Target expiration window: 5 to 45 DTE	GAMMA PIN GUARD
Gate 6	Drawdown Breaker	Immediate freeze if daily loss hits -3.0%	CIRCUIT BREAKER
Gate 7	Dynamic Sizing	Auto-scales contracts to stay inside \$2k cap	DYNAMIC SCALE

4. FastMCP Tools & Quantum Observability

Seamless agent tool integration and real-time execution telemetry

FastMCP Autonomous Agent Tools: Exposes 5 standalone MCP tools for AI IDEs (Cursor, Claude Desktop, Antigravity) — `get_account_telemetry`, `scan_and_propose_spread`, `validate_risk_gate`, `execute_options_order`, and `emergency_kill_switch`.

Quantum Dark Web Dashboard: Real-time paper trading interface featuring dynamic payoff curves, Greeks matrix (Delta/Gamma/Theta/Vega/IV), live orders and active positions tracking, and 1-click autonomous cycle execution.

Submission Links:

- **Web Dashboard:** <https://aegis-alpha-desk.vercel.app>
- **Interactive Slides:** <https://aegis-alpha-desk.vercel.app/slides.html>
- **GitHub Repo:** <https://github.com/DeathKnell837/band-of-agents-hackathon>